

# Ryan Peach

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**Objective:** Senior specialist in AI and Cloud Technologies seeking meaningful, impact-driven work.

## EDUCATION

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|-------------------------------------------------------------------------------------------------------------|-------------------|
| <b>Georgia Tech</b>                                                                                         | Atlanta, GA       |
| <i>MS in Computer Science, Machine Learning Specialty. Computer Vision &amp; Robotics Equiv. Coursework</i> |                   |
| 2015 – 2018                                                                                                 |                   |
| <b>Western Kentucky University</b>                                                                          | Bowling Green, KY |
| <i>BS in Electrical Engineering, Minor in Mathematics</i>                                                   |                   |
| 2010 – 2014                                                                                                 |                   |

## MAJOR SKILLS

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|----------------------------------------------------------------------------------------|-------------------------------|
| <i>Python Scientific Stack, Data Science</i>                                           | <i>Expert, 2013-Present</i>   |
| <i>Docker, Kubernetes, AWS</i>                                                         | <i>Expert, 2017-Present</i>   |
| <i>Machine Learning: Deep Learning, Reinforcement Learning</i>                         | <i>MS, 2015-Present</i>       |
| <i>Large Language Models: Safety, Architecture, API Tools, Open Source Development</i> | <i>Advanced, 2022-Present</i> |
| <i>Engineering: 3D Printing, PCB Design, OnShape</i>                                   | <i>BS, 2010-Present</i>       |
| <i>Rust Language</i>                                                                   | <i>Advanced, 2019-Present</i> |

## EXPERIENCE

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|----------------------------------|--------------------|
| <b>Senior Principal Engineer</b> | 2023-Present       |
| <i>Ansys / Synopsys</i>          | <i>Atlanta, GA</i> |
| <i>Architecture</i>              |                    |

- Trained management on transition from single team Kubernetes infrastructure to enterprise infrastructure.
- Architected globally available infrastructure, designed and maintained authoritative infrastructure diagrams covering approximately dozens of interoperating services.
- Designed GraphQL, gRPC, and OpenAPI APIs for core backend services.

### *Large Language Models*

- Participated in early open source efforts with LangChain, LlamaIndex, etc. Contributing code to repos.
- Advised on company LLM projects across multiple teams, attended conferences on LLM research

|                                 |                    |
|---------------------------------|--------------------|
| <b>Director of Cloud DevOps</b> | 2021-2023          |
| <i>OnScale</i>                  | <i>Atlanta, GA</i> |

### *Architecture*

- Managed custom Kubernetes controller which managed simulation jobs on high-performance compute (HPC) infrastructure.
- Refactored and modernized Infrastructure as Code (IaC).
- Managed AWS pricing, SEV0 outages, observability, and data analytics

### *Machine Learning*

- Managed team and cloud resources for development of ML Pipeline which predicted simulation time based on simulation metrics prior to runtime, giving customer an estimate of pricing.
- ML Pipeline needed to automatically retrain on new data periodically so architected ETL and SageMaker training

### *Management*

- Completely up and running within 2 weeks of hiring due to leadership transition.
- Directed all employees and decisions on Backend, Architecture, and ML projects.
- Managed ML, HPC, Data, and DevOps teams during Ansys acquisition.
- Transitioned to technical track after transitioning all teams to new management under Ansys.

## Machine Learning Researcher - Lead Architect

Keysight Technologies

2017 – 2021

Atlanta, GA

### Architecture

- Built a CloudFormation stack using ECR, Lambda that spins up thousands of Docker containers to train, tune, and select optimal SVG models. Weeks of compute completed in an hour regardless of scale.
- Leading the team in code reviews, managed all the DevOps for the team including managing the git repo and the Jenkins CI/CD.
- Designed PC cluster hardware and managed on-prem Unix, Spark, PostgreSQL, MongoDB, and Kubernetes server.

### Machine Learning

- Independently investigated almost a dozen potential ML applications at the company, providing preliminary results and feasibility studies.
- Wrote approximately 7 internal publications on patentable inventions, as well as giving multiple company-wide presentations on my work.
- Implemented large scale hyperparameter tuning systems on AWS. Some custom and others using SageMaker. Worked with AutoML solutions.
- Applied Reinforcement Learning to multiple novel problem domains using Monte Carlo Tree Search and Q Learning.
- Data visualization and web-enabled dashboard creation in Tensorboard, Plotly, Dash, Django, Tableau (similar software), Matplotlib, Altair, and Jupyter.

### Leadership

- Promoted from Developer to Researcher within only 1 year due to outstanding performance. Won top 10% performer every year with the company.
- Interviewed over a dozen candidates within 3 years.
- Onboarded 3 coworkers in Python best practices and professional software design principles.
- Maintained coding standards, code review, and planning of development milestones.

## TECHNICAL SKILLS

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**Primary Languages:** Python, Rust, TypeScript, Bash/Unix, PostgreSQL, Terraform, CloudFormation

**Additional Known Languages:** Golang, C/C++, C#, Java, Haskell, Clojure, Spark/Scala, Cypher

**Technologies:** Git, Docker, Kubernetes, Supabase, Redis, MongoDB, DynamoDB, Neo4j, GitHub Actions, AWS, Ansible, Datadog, Linux

## COMMUNITY

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### Open Source Contributions

- [run-llama/llama\\_index](#)
- [OpenHands/OpenHands](#)
- [guidance-ai/guidance](#)

### Open Source Creations (including Archives)

- [ryanpeach/continuing\\_education](#) (In Progress) - A Python repo where I track ML papers and do replications.
- [ryanpeach/mdlinker](#) (Archived) - Rust executable that runs at git pre-commit to detect and repairs broken wikilinks, duplicate files, and missing aliases in Obsidian and similar markdown note taking apps.
- [ryanpeach/orbitingsandrust](#) (Archived) - A game physics engine I worked on in Rust which I think demonstrates my Rust code quality.

### Local Activities

- [PyATL \(Python Atlanta\)](#) – Frequent Speaker on Machine Learning and Data Science in Python.